

**STABILITY FIELD OF ENTHALPICALLY STABILIZED COMPOUNDS
OF VARIABLE COMPOSITION WITH PYROCHLORE STRUCTURE
IN THE $\text{Bi}_2\text{O}_3\text{--Fe}_2\text{O}_3\text{--WO}_3\text{--(H}_2\text{O)}$ SYSTEM**

Lomakin M.S.^(1,2), Proskurina O.V.^(1,3)

⁽¹⁾ Ioffe Institute

194021, Saint Petersburg, Politekhnikeskaya st., 26

⁽²⁾ Institute of Silicate Chemistry

199034, Saint Petersburg, Makarov emb., 2

⁽³⁾ St. Petersburg State Institute of Technology

190013, Saint Petersburg, Moskovsky ave., 26

When using the hydrothermal synthesis method, the stabilization of the crystalline phase occurs at low temperatures (100 – 200 °C), which means that it is primarily due to the enthalpic contribution to the change in Gibbs energy ($\Delta G = \Delta H - T\Delta S$), while the influence of the entropic term is minimized.

Compounds of variable composition with pyrochlore structure in the $\text{Bi}_2\text{O}_3\text{--Fe}_2\text{O}_3\text{--WO}_3$ system, obtained under hydrothermal conditions ($T = 90 - 200$ °C, $P \leq 7$ MPa), are enthalpically stabilized phases [1], since they are unstable at temperatures above ~ 725 °C [2]. The study of these compounds is of particular interest because their properties may reveal features that are not typical of compounds in which the entropy factor plays a significant role in the change in Gibbs free energy.

In this work, the concentration limits of the stability field of $(\text{Bi}_z\text{Vac}_{2-z})(\text{Fe}_{2-y}\text{W}_y)\text{O}_6\text{O}'_\delta$ compounds (where Vac is a cation vacancy) with pyrochlore structure ($T = 200$ °C and $P \sim 7$ MPa: $0.60 \leq z \leq 1.35$; $1.32 \leq y \leq 1.96$; $\delta \sim 1$) were determined and it was shown that they are several times higher than those for pyrochlore phases obtained in most other oxide systems using the solid-phase synthesis method.

1. Fedorov P.P., Almyasheva O.V., Alexandrov A.A., Proydakova V.Yu., Korotkova N.A., Baranovskaya V.B., Gusarov V.V. Low-temperature phase formation in the $\text{ZrO}_2\text{--In}_2\text{O}_3$ system // Mendelev Comm. 2025. Vol. 35. P. 376–378. <https://doi.org/10.71267/mencom.7642>

2. Lomakin M.S., Proskurina O.V., Levin A.A., Kozlov S.S., Semenov V.G., Kozlov V.S., Koroleva A.V., Zhizhin E.V., Shevaleevskiy O.I. Pyrochlore phase in the $\text{Bi}_2\text{O}_3\text{--Na}_2\text{O--Fe}_2\text{O}_3\text{--WO}_3\text{--(H}_2\text{O)}$ system: crystal structure and energy band diagram depending on chemical composition // Ceram. Int. 2025. Vol. 51. P. 43488–43510. <https://doi.org/10.1016/j.ceramint.2025.06.323>

The authors express their deep gratitude to Corr. Mem. RAS Victor Vladimirovich Gusarov for valuable comments and advice on improving the quality of the research.

The work of M.S.L. was financially supported by the Russian Science Foundation (Project № 24-13-00445).